



Citation integrity in the age of AI: evaluating the risks of reference hallucination in maxillofacial literature

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ABSTRACT

The increasing adoption of large language models (LLMs) such as ChatGPT in academic writing has introduced both opportunities and risks. While these tools enhance productivity and accessibility, their reliability in generating accurate references remains uncertain. This short communication highlights the growing concern of 'reference hallucination', where AI-generated citations appear legitimate but are fabricated or contain significant metadata errors. Across all scientific disciplines, including oral and maxillofacial surgery (OMFS), where evidence-based practice is foundational, such inaccuracies can undermine academic integrity and clinical trust. This article summarizes common reference-related errors reported in literature and calls for heightened editorial vigilance, AI-literacy training, and the integration of real-time bibliographic tools. Responsible use of AI in scholarly publishing is essential to preserving the quality and credibility of surgical literature.

1. Short communication

The introduction of generative artificial intelligence (AI) tools, particularly large language models (LLMs) such as ChatGPT, has rapidly transformed academic writing practices across medical and surgical specialties. These models are increasingly used to generate drafts, summarize articles, and assist in manuscript preparation (Singhal et al., 2023; Kung et al., 2023). While these technologies offer significant advantages in speed and accessibility, their reliability — especially in producing accurate and verifiable references — has come under scrutiny (Alkaissi and McFarlane, 2023).

While reference accuracy is essential across all scientific and medical disciplines, its importance becomes especially apparent in fields like oral and maxillofacial surgery (OMFS), where clinical decisions are closely tied to current evidence. However, studies have highlighted a concerning trend: ChatGPT and other LLMs can generate citations that appear plausible but are in fact fabricated or contain critical metadata errors (Chelli et al., 2024; Gao et al., 2023). These 'hallucinated references' may contain fictitious article titles, incorrect author names, or DOIs that do not resolve. Such inaccuracies pose a significant threat to academic integrity and, if left undetected, could lead to the dissemination of

misleading information.

The core limitation stems from the architecture of LLMs. These systems are trained on vast textual datasets, but lack live integration with bibliographic databases like PubMed or CrossRef. Their outputs are based on linguistic probability rather than factual retrieval, increasing the risk of reference hallucination when generating academic content. It is important to note that this phenomenon is not exclusive to OMFS, but has been observed across diverse domains, including internal medicine, psychology, and public health literature.

Major academic publishers have acknowledged this risk. Journals such as JAMA have updated editorial policies to mandate disclosure of AI usage and to discourage reliance on AI-generated citations unless manually verified (Flanagin et al., 2023). However, policy alone may not suffice. Editorial boards and reviewers must adopt automated tools to detect reference inaccuracies and promote AI-literacy training among researchers, especially early-career scholars.

In conclusion, while LLMs like ChatGPT hold promise as productivity tools, their limitations in citation fidelity necessitate human oversight. In all fields of science — and particularly in clinical disciplines like OMFS, where evidence-based practice is vital — references should be cross-verified and not accepted at face value. Until integration with real-

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Table 1
Common types of reference error in ChatGPT-generated citations (based on published studies).

Type of error	Description	Reported in
Fabricated references	Entirely non-existent articles with a plausible structure	Alkaissi and McFarlane (2023); Chelli et al. (2024)
Incorrect metadata	Real articles with wrong author names, DOIs, or journal information	Gao et al. (2023); Chelli et al. (2024)
Recycled author names	Known researchers paired with false titles or journals	Alkaissi and McFarlane (2023)
Non-resolving DOIs	DOI numbers that look valid but do not link to real publications	Chelli et al. (2024)
Journal title inflation	Assigning false articles to high-impact journals	Gao et al. (2023)

time bibliographic databases is achieved, responsible use of AI in scientific writing must remain a priority.

Table 1 summarizes common types of reference error reported in recent literature evaluating ChatGPT’s citation performance (Alkaissi and McFarlane, 2023; Chelli et al., 2024; Gao et al., 2023).

To mitigate reference hallucination, researchers may consider using AI-integrated bibliographic tools such as Scite.ai, Elicit.org, Research Rabbit, and Consensus.app, which can aid in retrieving source-verified citations. While these platforms do not fully eliminate the need for manual verification, they offer a more reliable alternative to standard LLMs by integrating evidence scoring and citation tracing. Future AI developments that directly integrate with databases like PubMed or CrossRef may further reduce citation errors. Until then, human oversight remains indispensable (Aljamaan et al., 2024).

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Conflicts of interest

The authors declare no conflict of interest.

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